

Three Graffiti.pc Conjectures on Largest Induced Trees: Proofs of Conjectures 141, 142, and 143

Alper Ferudun
alper@mercurycodelab.com

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Abstract

For a finite simple graph G , let $t(G)$ be the largest order of an induced tree and let $g(G)$ be the girth. We prove three consecutive conjectures of DeLaViña's Graffiti.pc program. First, writing $\ell(v)$ for the independence number of the subgraph induced by the neighbourhood of v , we prove

$$t(G) \geq \left\lfloor \frac{g(G)}{2} \right\rfloor - 1 + \max_{v \in V(G)} \ell(v).$$

Second, if $\text{Per}(G)$ is the periphery and $f(G) = \max_x d(x, \text{Per}(G))$, we prove

$$t(G) \geq \frac{2}{3} g(G) + f(G),$$

and establish the stronger integral bound $t(G) \geq f(G) + \lceil 2g(G)/3 \rceil$ when G contains a cycle. Third, if $\delta'(G)$ is the second-smallest degree, counted with multiplicity, then every connected non-tree graph satisfies

$$t(G) \delta'(G) \geq g(G) + 1.$$

These are Conjectures 141, 142, and 143 of *Written on the Wall II*. Complete, machine-checked Lean 4 proofs of all three formal statements accompany the manuscript.

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1 Overview

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz’s Graffiti program [5], generates conjectural inequalities between graph invariants. Its conjectures are collected in *Written on the Wall II* [3]; a curated register is maintained by West [4]. The three consecutive entries treated here concern the maximum order of an induced tree, whose systematic study goes back to Erdős, Saks, and Sós [6].

The three proofs are logically independent and are presented in self-contained sections. Conjecture 141 follows from a stronger maximum-degree–girth bound for triangle-free cyclic graphs. Conjecture 142 is driven by a rooted shortest-cycle construction and a three-point metric estimate. Conjecture 143 reduces to a two-leaf induced-tree lemma. The associated Lean developments are recorded in Formal Conjectures pull requests #4454 and #4457 [15, 16] and are included as ancillary files.

Use of generative AI. OpenAI GPT Pro and Codex were used for proof exploration, literature-search assistance, computational checks, Lean development, and manuscript preparation. They are not authors. Responsibility for verifying the work and for the submitted version rests entirely with the named human author.

2 Conjecture 141: neighbourhood independence

2.1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz’s program Graffiti [5], generates conjectural inequalities between graph invariants; its conjectures are collected in the lists *Written on the Wall II* [3], with a curated register maintained by West [4]. A run of these conjectures bounds from below the invariant $t(G)$, the largest order of an induced subtree, whose systematic study goes back to Erdős, Saks, and Sós [6].

Throughout, G is a finite simple graph, $g(G)$ its girth, $\Delta(G)$ its maximum degree, and for a vertex v we write $\ell(v)$ for the independence number of the subgraph induced by the neighbourhood $N(v)$. Conjecture 141 of *Written on the Wall II*, dated 2005, asserts [3]:

If G is a simple connected graph, then $t(G) \geq \lfloor g(G)/2 \rfloor - 1 + \max_v \ell(v)$.

At the time of writing the conjecture is listed as open on DeLaViña’s register (status “O”) [3] and is stated as a research-open item in the Google DeepMind *Formal Conjectures* repository [9,

[11]. We prove it. The main step is the following sharper bound, which does not involve ℓ at all and may be of independent interest.

Theorem 2.1. *Let G be a finite simple connected triangle-free graph that contains a cycle. Then*

$$t(G) \geq \Delta(G) + g(G) - 3.$$

Equality holds for every cycle C_g ($g \geq 4$) and every complete bipartite graph $K_{a,b}$ with $a, b \geq 2$.

Corollary 2.2 (Conjecture 141). *Every finite simple connected graph G on at least two vertices satisfies $t(G) \geq \lfloor g(G)/2 \rfloor - 1 + \max_v \ell(v)$, where $g(G) := 0$ if G is acyclic.*

The deduction of the corollary is short: for girth ≤ 5 (and for acyclic graphs) an induced star at a vertex maximizing ℓ already beats the bound, while for girth ≥ 6 the graph is triangle-free, so $\ell(v) = \deg(v)$ for every v , and Theorem 2.1 applies with $g - 3 \geq \lfloor g/2 \rfloor - 1$.

The proof of Theorem 2.1 is a maximality argument in the spirit of our proof of Graffiti.pc’s Conjecture 143 [15]: a maximum-order induced tree containing the closed neighbourhood of a maximum-degree vertex either spans (impossible in a cyclic graph) or admits an outside vertex with two neighbours on the tree; the tree path joining those two neighbours closes a cycle, hence has length at least $g(G) - 2$, and it can share at most three vertices with the closed neighbourhood.

Related work. The foundational paper [6] bounds $t(G)$ in terms of order, size, radius, and independence and clique numbers; we are not aware of a published lower bound for t combining girth with the maximum degree or with neighbourhood independence, and the closest Graffiti.pc antecedents (the induced-forest bounds of DeLaViña–Waller [7]; see also Hertz–Marcotte–Schindl [8]) concern induced forests rather than trees. DeLaViña’s resolved-conjecture lists do not contain Conjecture 141 [3]. As with any short elementary argument, we do not claim it could not have been observed before.

Verification. Theorem 2.1 and Corollary 2.2 have been formalized and machine-checked in Lean 4 [18] on top of Mathlib [19], against the pre-existing formal statement `conjecture141` of the Formal Conjectures repository [11]; see Section 2.6. Independently, the conjecture and all intermediate claims were verified exhaustively by exact computation on all 995 connected graphs on 2–7 vertices and on structured families (cages, prisms, generalized theta graphs, cycles with pendant trees, complete bipartite graphs, and seeded random graphs); no violation was found, and every equality case located has girth exactly 4. The computations play no role in the proofs.

2.2 Notation

All graphs are finite and simple. For $S \subseteq V(G)$, $G[S]$ is the induced subgraph; S *induces a tree* if $G[S]$ is connected and acyclic, and $t(G) = \max\{|S| : G[S] \text{ is a tree}\}$. The girth of a graph containing a cycle is the minimum length of a cycle; following the Mathlib convention used by the formal statement, an acyclic graph has girth 0. $N(v)$ and $N[v] = N(v) \cup \{v\}$ are the open and closed neighbourhoods; $\ell(v)$ is the independence number of $G[N(v)]$. Note $\ell(v) \leq \deg(v)$ always, with equality when $N(v)$ is independent, i.e. for all v when G is triangle-free.

2.3 The star lemma

Lemma 2.3. *For every vertex v of any graph G : $t(G) \geq \ell(v) + 1$.*

Proof. Let $S \subseteq N(v)$ be independent with $|S| = \ell(v)$. Then $G[\{v\} \cup S]$ is a star: all edges vs ($s \in S$) are present and there are no edges inside S . A star is a tree. $\square$

2.4 Proof of Theorem 2.1

Proof of Theorem 2.1. Write $g = g(G) \geq 4$ and $\Delta = \Delta(G)$. Fix a vertex v of degree Δ . Since G is triangle-free, $N(v)$ is independent, so by the proof of Lemma 2.3 the closed neighbourhood $N[v]$ induces a star, hence a tree. Among all sets $S \supseteq N[v]$ inducing a tree choose one, say S^* , of maximum cardinality, and let $T = G[S^*]$.

S^* is a proper subset of $V(G)$: an induced spanning tree would make G itself a tree, contradicting the existence of a cycle. Since G is connected, some vertex $z \notin S^*$ has a neighbour in S^* ; and z must have two distinct neighbours $a, b \in S^*$, for otherwise $G[S^* \cup \{z\}]$ would be the tree T with a pendant vertex attached, an induced tree containing $N[v]$ and larger than T — contradicting maximality.

Let P be the unique a – b path in T . The edges of P together with za and zb form a cycle of length $|E(P)| + 2$, so

$$|E(P)| \geq g - 2. \quad (2.1)$$

Next, $|V(P) \cap N[v]| \leq 3$. Every edge of G between two vertices of S^* is an edge of the tree T (as T is induced), and paths between fixed endpoints in a tree are unique. If $v \notin V(P)$ and P contained two distinct vertices $u_1, u_2 \in N(v)$, then the segment of P between u_1 and u_2 would be the unique u_1 – u_2 path in T ; but u_1vu_2 is also a u_1 – u_2 path in T , so the segment would pass through v , contradicting $v \notin V(P)$; hence $|V(P) \cap N[v]| \leq 1$ in this case. If $v \in V(P)$, then for any $u \in V(P) \cap N(v)$ the segment of P between v and u is the unique v – u path of T , which is the single edge vu ; so u is adjacent to v along P , and a path has at most two edges at any vertex: $|V(P) \cap N(v)| \leq 2$, so $|V(P) \cap N[v]| \leq 3$.

Finally, count. $V(P)$ and $N[v]$ are subsets of S^* , so by inclusion–exclusion and (2.1),

$$|S^*| \geq |V(P) \cup N[v]| = |V(P)| + |N[v]| - |V(P) \cap N[v]| \geq (g - 1) + (\Delta + 1) - 3 = \Delta + g - 3.$$

Since T is an induced tree, $t(G) \geq |S^*| \geq \Delta + g - 3$.

For sharpness: in C_g ($\Delta = 2$) the largest induced trees are the paths obtained by deleting one vertex, so $t = g - 1 = \Delta + g - 3$; in $K_{a,b}$ with $a \leq b$ ($g = 4$, $\Delta = b$) every induced tree is a star (two vertices on each side already induce C_4), so $t = b + 1 = \Delta + g - 3$. $\square$

Remark 2.4. Under the hypotheses of Theorem 2.1, an alternative proof of the slightly weaker bound $t \geq \Delta + \lfloor g/2 \rfloor - 1$ (still sufficient for Corollary 2.2) is worth recording. For $r = \lfloor g/2 \rfloor - 1$, the ball $B(v, r)$ induces a tree in any graph of girth g : every G -edge inside the ball is an edge of a breadth-first search tree rooted at v , since a non-tree edge would close a cycle of length at most $2r + 1 < g$ through the last common ancestor of its endpoints. Now take v of maximum degree; triangle-freeness gives $g \geq 4$, so $r \geq 1$. If G contains a cycle the ball is proper, so all r distance layers are nonempty and $|B(v, r)| \geq 1 + \Delta + (r - 1)$.

2.5 Proof of Corollary 2.2

Proof of Corollary 2.2. Let $L = \max_v \ell(v)$; note $L \geq 1$ since G is connected on ≥ 2 vertices. If G is acyclic then $g = 0$ and the claimed bound is $L - 1 \leq t$, which follows from Lemma 2.3. If $3 \leq g \leq 5$ then $\lfloor g/2 \rfloor - 1 \leq 1$ and Lemma 2.3 again gives $t \geq L + 1 \geq \lfloor g/2 \rfloor - 1 + L$. If $g \geq 6$ then G is triangle-free, so $\ell(v) = \deg(v)$ for every v and $L = \Delta$; by Theorem 2.1, $t \geq \Delta + g - 3 \geq \Delta + \lfloor g/2 \rfloor - 1 = L + \lfloor g/2 \rfloor - 1$, using $g - 3 \geq \lfloor g/2 \rfloor - 1$ for $g \geq 4$. $\square$

Remark 2.5. The corollary is sharp precisely in girth 4. Equality holds for C_4 , for every $K_{a,b}$ ($a, b \geq 2$), and for the infinite family obtained from C_4 by attaching $k \geq 0$ pendant vertices to

one vertex ($\ell_{\max} = k + 2$, $t = k + 3$); an exhaustive check of all connected graphs on at most seven vertices finds no equality case of any other girth. That equality *requires* girth 4 follows from the results above: for acyclic G and for $g = 3$ the star bound $t \geq L + 1$ beats the right-hand side by at least 2 resp. 1; for $g = 5$ triangle-freeness gives $L = \Delta$ and Theorem 2.1 yields $t \geq \Delta + 2 = L + \lfloor 5/2 \rfloor$, one more than required; and for $g \geq 6$ the inequality $g - 3 > \lfloor g/2 \rfloor - 1$ is strict.

2.6 Formalization

The Formal Conjectures project [9] maintains Lean 4 formalizations of open conjectures. Conjecture 141 appears in `GraphConjecture141.lean` as `conjecture141`, marked `research open` [11], in the denominator-explicit integer form

$$\lfloor g(G)/2 \rfloor - 1 + \max_v \ell(v) \leq t(G).$$

Here t and ℓ are the repository definitions `largestInducedTreeSize` and `indepNeighborsCard`; Mathlib girth is natural-valued and equals zero on acyclic graphs.

We have produced a complete, `sorry`-free Lean 4 proof of exactly this statement, machine-checked with Lean toolchain v4.27.0 against current Mathlib, together with the supporting API: the star construction, the maximum-induced-tree selection with prescribed vertices, the two-neighbour maximality obstruction, the tree-path girth certificate, and the three-vertex overlap bound of Theorem 2.1. The development builds on the reusable induced-tree API we contributed alongside Conjecture 143 [15]. The proof uses no `native_decide` and no additional axioms (`#print axioms` reports `propext`, `Classical.choice`, `Quot.sound`). The source is available as pull request #4454 to the Formal Conjectures repository (which also contains our formalization of Conjecture 143) and as ancillary files with this submission.

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3 Conjecture 142: distance from the periphery

3.1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz’s Graffiti program [5], generates conjectural inequalities among graph invariants. Its conjectures are collected in *Written on the Wall II* [3]; a curated selection is maintained by West [4]. Several consecutive entries concern the maximum order of an induced tree, an invariant systematically studied by Erdős, Saks, and Sós [6].

Let G be a finite simple connected graph. The *periphery* $\text{Per}(G)$ is the set of vertices of eccentricity $\text{diam}(G)$, and the set-eccentricity of the periphery is

$$f(G) := \max_{x \in V(G)} d(x, \text{Per}(G)).$$

Conjecture 142 of *Written on the Wall II* asserts

$$t(G) \geq \frac{2}{3} g(G) + f(G).$$

Here and in the formal statement, the girth of an acyclic graph is taken to be zero. We prove the following stronger cyclic statement.

Theorem 3.1 (Graffiti.pc Conjecture 142). *Let G be a finite simple connected graph containing a cycle. If $g = g(G)$ and $f = f(G)$, then*

$$t(G) \geq f + \left\lceil \frac{2g}{3} \right\rceil.$$

Consequently every finite simple connected graph satisfies $t(G) \geq \frac{2}{3}g(G) + f(G)$.

The proof is self-contained apart from elementary facts about finite graphs. Its structural core is Lemma 3.6, which turns three metric terminals into an admissible forest attached to a shortest cycle. The resulting metric inequality, Lemma 3.7, is then combined with a short classification of the equality case in which the admissible forest has only one vertex.

3.2 Preliminaries

Write $D = \text{diam}(G)$ and $B = \text{Per}(G)$. A shortest path is induced, so

$$t(G) \geq D + 1. \quad (3.1)$$

A shortest cycle K is chordless and isometric: for $u, v \in V(K)$, their graph distance equals their shorter arc-distance in K . Deleting one vertex of K therefore gives

$$t(G) \geq g - 1. \quad (3.2)$$

If $f \geq 1$, then

$$D \geq f + 1. \quad (3.3)$$

Indeed, always $f \leq D$; equality would give a vertex x at distance D from every peripheral vertex, hence $\text{ecc}(x) = D$ and $x \in B$, contradicting $f = d(x, B) \geq 1$.

For a shortest cycle K , define $M(K)$ to be the maximum of $|F|$ over all sets $F \subseteq V(G) \setminus V(K)$ for which $G[F]$ is a forest and there is a vertex $z \in V(K)$ such that every component C of $G[F]$ sends exactly one edge into $V(K) \setminus \{z\}$. Edges from C to z are unrestricted.

Lemma 3.2 (Cycle–forest extension). *For every shortest cycle K ,*

$$t(G) \geq g - 1 + M(K).$$

Proof. Choose F, z realizing $M(K)$ and let c be the number of components of $G[F]$. The graph induced by $(V(K) \setminus \{z\}) \cup F$ is connected. Its edges consist of the $g - 2$ edges of the path $K - z$, the $|F| - c$ forest edges, and one attachment edge for each of the c components. Thus it has $g - 1 + |F|$ vertices and one fewer edge, and hence is a tree. $\square$

Lemma 3.3 (An extra cycle vertex). *Suppose $f \geq 1$. If $g \geq 4$, then $t(G) \geq g$; if $g \geq 5$, then $M(K) \geq 1$ for every shortest cycle K .*

Proof. If $V(G) = V(K)$, chordlessness and connectedness give $G = C_g$, whose every vertex is peripheral, contrary to $f \geq 1$. Hence some vertex $y \notin K$ is adjacent to K . When $g \geq 5$, it has exactly one neighbour on K : two neighbours and the shorter arc between them would form a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$. Taking $F = \{y\}$ and deleting a cycle vertex other than its root proves both assertions. For $g = 4$, the only additional possibility is that y has two antipodal neighbours on K ; deleting one of them again leaves an induced tree on four vertices. $\square$

We shall also need two elementary connector lemmas.

Lemma 3.4 (Three terminals). *Every three distinct vertices of a connected triangle-free graph lie in a common induced tree.*

Proof. Let P be an a - b geodesic. If $c \in V(P)$, then P itself is the required induced tree. Otherwise, let $R = r_0 \cdots r_\ell$ be a shortest path from $c = r_0$ to P . Put $q = r_{\ell-1}$ and $R^\circ = R - r_\ell$. Only q can have additional neighbours on P . Any two such neighbours have P -distance at most two, and triangle-freeness excludes distance one; hence q has at most two neighbours on P . With one neighbour, $P \cup R^\circ$ induces a tree. With two neighbours u, v , their P -subpath is umv ; deleting m joins the two components of $P - m$ through q and yields a connected induced graph with one fewer edge than vertices. Since $m \notin \{a, b\}$, this is the required tree. $\square$

Lemma 3.5 (Cycles in a minimal connector). *Let J be connected and $Q \subseteq V(J)$. If no proper induced connected subgraph of J contains Q , then every cycle L of J has $|V(L)| \leq |Q|$.*

Proof. For every $q \in Q$, choose a shortest path from q to L , and denote its unique first vertex on L by $c(q)$. Fix $v \in L$. The path $L - v$ lies in one component J_v of $J - v$. Minimality implies that some $q_v \in Q$ is outside J_v . Its chosen path to L must therefore end at v , so $c(q_v) = v$. Distinct vertices of L give distinct terminals, which injects $V(L)$ into Q . $\square$

3.3 A rooted shortest-cycle lemma

Lemma 3.6 (Rooted shortest cycle). *Assume $g \geq 5$, and let $S \subseteq V(G)$ with $|S| \leq 3$. There exist a shortest cycle K , a vertex $z \in V(K) \setminus S$, and a set $F \subseteq V(G) \setminus V(K)$ such that*

$$S \subseteq (V(K) \setminus \{z\}) \cup F,$$

$G[F]$ is a forest, and every component C of $G[F]$ satisfies

$$e(C, V(K) \setminus \{z\}) = 1.$$

Proof. The empty case is immediate. Fix an arbitrary shortest cycle K , and choose $X \supseteq V(K) \cup S$ of minimum cardinality such that $H := G[X]$ is connected.

Every component C of $H - V(K)$ contains a terminal. Indeed, it has an edge to K , since H is connected; if $S_C := S \cap V(C)$ were empty, the whole component could be deleted while preserving connectedness and all required vertices. Put

$$A_C := \{a \in V(C) : N_G(a) \cap V(K) \neq \emptyset\}.$$

Form J_C from $G[C]$ by adjoining a new vertex r_C adjacent precisely to A_C . This graph is vertex-minimal among induced connected subgraphs containing $S_C \cup \{r_C\}$. Otherwise, replacing C by a smaller connector would leave every resulting component attached to K and would contradict the choice of X .

Lemma 3.5 shows that every cycle in J_C has length at most $|S_C| + 1 \leq 4$. A cycle contained in $G[C]$ would also be a cycle of G and would have length at least $g \geq 5$. Consequently

$$G[C] \text{ is a tree.} \tag{3.4}$$

Let R_C be the minimal subtree of C containing A_C . We claim

$$|V(R_C)| \leq |S_C|. \tag{3.5}$$

For $v \in V(R_C)$, let Q_v be the component of $J_C - v$ containing r_C . Minimality supplies $s_v \in S_C \setminus V(Q_v)$. For $s \in S_C$, let $c(s)$ be the first vertex of R_C on the unique path from s to R_C . Every nonempty component of $R_C - v$ contains a vertex of A_C , by the minimality of R_C . Thus, if $v \neq c(s)$, the path from s to $c(s)$ and then to such an attachment gives an s - r_C path in $J_C - v$; hence $s \in Q_v$. Applying this to s_v gives $c(s_v) = v$. The map $v \mapsto s_v$ is injective, proving (3.5).

Every $a \in A_C$ has exactly one neighbour on K . Existence follows from the definition, while two distinct neighbours and their shorter K -arc would form a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$. Denote the unique root by $\rho(a)$. If $a, a' \in A_C$ are distinct, then

$$d_C(a, a') \leq |V(R_C)| - 1 \leq 2 \quad (3.6)$$

and their roots are distinct: equal roots, together with the a - a' path in C , would form a cycle of length at most four. Moreover, the two root edges, that path, and a shorter arc of K give

$$d_K(\rho(a), \rho(a')) + d_C(a, a') + 2 \geq g. \quad (3.7)$$

These inequalities sharply restrict the number of attachments. If $g \geq 9$, then two attachments would make the left side of (3.7) at most $\lfloor g/2 \rfloor + 4 < g$, so there is only one. For $g \in \{7, 8\}$, two attachments cannot be adjacent in C ; three attachments would fill the three vertices of R_C and contain an adjacent pair. For $g = 6$, if $R_C = p - q - r$ consisted of three attachments, applying (3.7) to p, q and to q, r would force both $\rho(p)$ and $\rho(r)$ to be the unique antipode of $\rho(q)$ on the 6-cycle, contrary to distinctness. Thus every component has at most two attachments, except that for $g = 5$ one component may have three. Also, at most one component has two or more attachments, because such a component contains at least two terminals and $|S| \leq 3$.

Suppose first that every component has one attachment a_C . The set

$$Q = (S \cap V(K)) \cup \{\rho(a_C) : C \text{ a component of } H - V(K)\}$$

has cardinality at most $|S| \leq 3$. Since $g \geq 5$, choose $z \in V(K) \setminus Q$ and take $F = X \setminus V(K)$. Equation (3.4) gives a forest, each component has its single attachment in Q , and $z \notin S$.

Next suppose that one component C_0 has exactly two attachments with roots u, v . It contains at least two terminals, so the set

$$Q = (S \cap V(K)) \cup \{\rho(a_C) : C \neq C_0\}$$

has size at most one. Choose $z \in \{u, v\} \setminus Q$ and again put $F = X \setminus V(K)$. Every ordinary component has one edge into $K - z$, while C_0 has exactly the two root edges and loses exactly one of them. Again $z \notin S$.

It remains to handle three attachments. Necessarily $g = 5$, $|V(R_C)| = |S_C| = |S| = 3$, there are no other off-cycle components, and $R_C = p - q - r$ with all three vertices attachments. Write

$$u = \rho(p), \quad v = \rho(q), \quad w = \rho(r).$$

Equation (3.7) gives $d_K(u, v) = d_K(v, w) = 2$. In cyclic order, write $K = u\alpha v\beta wu$. Then

$$K' = u\alpha vqpu$$

is another shortest 5-cycle.

The injection used to prove (3.5) is now a bijection. Hence there are distinct terminals s_p, s_q, s_r with $c(s_t) = t$. Let L_t be the unique t - s_t path in the tree C , and set

$$F' = (V(L_p) \setminus \{p\}) \cup (V(L_q) \setminus \{q\}) \cup V(L_r), \quad z = \alpha.$$

The three displayed path-pieces are pairwise disjoint and anticomplete, for an edge between two of them would create a cycle in the tree C . The first two nonempty pieces attach once to K' at p and q , respectively. The third attaches once through the edge rq ; its other old-cycle edge rw does not meet K' . Thus F' is admissible for (K', z) . Finally each s_t belongs either to its path-piece or, when $s_p = p$ or $s_q = q$, directly to $K' - z$. This completes the exceptional case and the proof. $\square$

3.4 The three-point bridge

Let

$$\mu := \max\{M(K) : K \text{ is a shortest cycle of } G\}.$$

Lemma 3.7 (Three-point inequality). *If $g \geq 5$, then every three distinct vertices a, b, c satisfy*

$$d(a, b) + d(b, c) + d(c, a) \leq g + 2\mu. \quad (3.8)$$

Proof. Apply Lemma 3.6 to $S = \{a, b, c\}$ and obtain K, z, F . Retain all edges of K , all edges inside the components of $G[F]$, and each component's unique edge into $K - z$; omit possible edges to z . The resulting spanning subgraph U of $G[K \cup F]$ is a single cycle K with rooted trees attached. It has exactly $|F|$ edges outside K : a component with k vertices contributes $k - 1$ internal edges and one root edge.

For each pair of terminals, use their unique paths to their roots and a shortest arc of K between the roots. An off-cycle edge separates from its root q of the three terminals. If its rooted tree contains h terminals, then its total multiplicity in the three selected paths is

$$q(h - q) + q(3 - h) = q(3 - q) \leq 2.$$

Thus the total off-cycle contribution is at most $2|F|$.

The sum of the three pairwise distances between any three roots on a cycle of length g is at most g . Indeed, if the roots are distinct and the three intervening arc-lengths are all at most $g/2$, their sum is exactly g ; if one arc exceeds $g/2$, the other two provide the shorter route for its endpoints and the sum is smaller. Coincident roots are immediate. Hence the three chosen paths have total length at most $g + 2|F| \leq g + 2M(K) \leq g + 2\mu$. Graph distances can only be shorter. $\square$

Corollary 3.8. *Assume $g \geq 5$ and $f \geq 1$. Then*

$$2\mu + g \geq 3f + 1. \quad (3.9)$$

Proof. Choose x with $d(x, B) = f$ and a diametral pair b, w . Both b, w are peripheral, so $d(x, b), d(x, w) \geq f$ and $d(b, w) = D \geq f + 1$ by (3.3). Apply Lemma 3.7 to x, b, w . $\square$

The estimate is almost sufficient by itself. The only delicate integer case is disposed of by the following sharp classification.

Lemma 3.9 (The case $\mu = 1$). *Assume $g \geq 5$ and $f \geq 1$. If $\mu = 1$, then G is obtained from C_g by attaching one pendant vertex to one cycle vertex. Consequently*

$$f \leq \left\lfloor \frac{g+2}{4} \right\rfloor. \quad (3.10)$$

Proof. Fix a shortest cycle K . Since $M(K) \leq \mu = 1$, no vertex can have distance two from K : the first two off-cycle vertices on a geodesic to K would form an admissible two-vertex path. Thus every vertex outside K is adjacent to K , and by $g \geq 5$ it has a unique root.

There cannot be two such vertices u, v . If they are nonadjacent, choose z different from both roots; the two singleton components are admissible. If they are adjacent, their roots are distinct (otherwise there is a triangle), and choosing z to be one root makes $\{u, v\}$ a connected admissible component. Either way $M(K) \geq 2$, a contradiction. Since $f \geq 1$, there is exactly one outside vertex, and G is the stated tadpole.

Put $k = \lfloor g/2 \rfloor$, let y be the pendant vertex, and let v_0 be its root. The diameter is $k + 1$; the periphery consists of y and the cycle vertices at cycle-distance k from v_0 . Every cycle vertex at position i on one of the v_0 -antipode arcs is at distance at most

$$\min(i + 1, k - i) \leq \left\lfloor \frac{k + 1}{2} \right\rfloor = \left\lfloor \frac{g + 2}{4} \right\rfloor$$

from the displayed peripheral vertices. The pendant vertex itself is peripheral, proving (3.10). $\square$

3.5 Completion of the proof

Proposition 3.10. *Theorem 3.1 holds when $g \geq 5$ and $f \geq 1$.*

Proof. Put $a = \lfloor g/3 \rfloor$. Since $\lceil 2g/3 \rceil = g - a$, the target is

$$t(G) \geq f + g - a. \quad (3.11)$$

If $f \leq a - 1$, the cycle bound (3.2) gives $t(G) \geq g - 1 \geq f + g - a$. Hence write $f = a + s$ with $s \geq 0$.

Lemma 3.3 gives $\mu \geq 1$, while Lemma 3.2 gives $t(G) \geq g - 1 + \mu$. It therefore suffices to prove

$$\mu \geq s + 1. \quad (3.12)$$

Corollary 3.8 and the three residues of g give:

$$\begin{array}{ll} g = 3a : & 2\mu \geq 3s + 1, \\ g = 3a + 1 : & 2\mu \geq 3s, \\ g = 3a + 2 : & 2\mu \geq 3s - 1. \end{array}$$

In the first case, $\mu \leq s$ would imply $2\mu \leq 2s < 3s + 1$. In the second, $s = 0$ is covered by $\mu \geq 1$, while for $s \geq 1$ we have $2s < 3s$. In the third, $s = 0$ is again immediate, and $s \geq 2$ gives $2s < 3s - 1$.

The only remaining possibility is $g = 3a + 2$ and $s = 1$. If $\mu = 1$, Lemma 3.9 yields

$$f \leq \left\lfloor \frac{3a + 4}{4} \right\rfloor \leq a,$$

because $a \geq 1$, contradicting $f = a + 1$. Hence $\mu \geq 2 = s + 1$. This proves (3.12), and therefore (3.11). $\square$

Proposition 3.11. *Theorem 3.1 holds for $g = 3$ and $g = 4$ when $f \geq 1$.*

Proof. For $g = 3$, equations (3.1) and (3.3) give $t(G) \geq D + 1 \geq f + 2 = f + \lceil 2g/3 \rceil$.

Let $g = 4$. Choose an f -realizer x and a diametral pair b, w . The graph is triangle-free, so Lemma 3.4 supplies an induced tree T containing x, b, w . Let T_0 be the minimal subtree containing them. Every edge of T_0 lies on exactly two of the three terminal-to-terminal paths; therefore

$$2|E(T_0)| = d_T(x, b) + d_T(x, w) + d_T(b, w) \geq 2f + D \geq 3f + 1.$$

If $f \geq 2$, then

$$|V(T)| \geq 1 + \left\lceil \frac{3f+1}{2} \right\rceil \geq f + 3.$$

If $f = 1$, Lemma 3.3 gives $t(G) \geq g = 4 = f + 3$. $\square$

Proof of Theorem 3.1. The preceding two propositions cover cyclic graphs with $f \geq 1$. If G is cyclic and $f = 0$, then $g - 1 \geq \lceil 2g/3 \rceil$ for $g \geq 3$, so (3.2) applies. This proves the integral cyclic assertion.

If G is acyclic, connectedness makes G itself a tree, so $t(G) = |V(G)|$ and $f \leq D \leq |V(G)| - 1$ (with the one-vertex case immediate). Since $g(G) = 0$, the real-valued Formal Conjectures inequality also follows. Finally, $\lceil 2g/3 \rceil \geq 2g/3$, completing the proof. $\square$

3.6 Verification and formalization status

The proof above was checked in two independent forms. First, a separate constructive proof based on descents to an arbitrary shortest cycle was implemented literally: every branch produces its cycle vertex z , its admissible forest, or its induced path, and a fresh checker verifies inducedness, connectedness, acyclicity, edge multiplicities, and the claimed cardinality. On a corpus of 10,776 connected cyclic graphs (the complete NetworkX atlas through seven vertices, structured extremal families, and seeded random and adversarial graphs through 32 vertices), the validator reported no failure. Repeating every free choice with two independent random seeds again gave no failure, for 32,328 checked certificates in all. These computations are supporting checks, not a substitute for the proof.

The exact statement of Conjecture 142 is formalized in the Google DeepMind *Formal Conjectures* repository [9, 12], using the repository’s largest-induced-tree invariant, Mathlib’s natural-valued girth, and the set-eccentricity of the boundary (peripheral) vertices. We have produced a complete, `sorry`-free Lean 4 proof of that statement. The proof is contained in `GraphConjecture142Proof.lean` at [commit 46bf390](#); the upstream statement is marked solved and imports this proof in pull request #4457 [16]. The development was compiled against the repository’s pinned toolchain and contains no `sorry` or `admit`. Both the proof file and the updated statement file are included with this submission as ancillary files.

Remark 3.12 (Sharpness). Equality occurs throughout the $g = 3$ diameter branch. Among the exhaustively checked small graphs, the only equality class of girth greater than three is the 6-cycle with one pendant vertex: here $f = 2$ and $t(G) = 6 = f + \lceil 2g/3 \rceil$. In the proof this is exactly the $\mu = 1$ tadpole case of Lemma 3.9.

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4 Conjecture 143: the second-smallest degree

4.1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz’s program Graffiti [5], generates conjectural inequalities between graph invariants. Its conjectures are collected in the lists *Written on the Wall II* [3]; a curated register with commentary is maintained by West [4]. Many of these conjectures concern the invariant $t(G)$, the largest order of an induced subgraph of G that is a tree, an invariant whose systematic study goes back to Erdős, Saks, and Sós [6].

Throughout, G is a finite simple graph. We write $g(G)$ for the girth of G (the length of a shortest cycle), and $\delta'(G)$ for the *second-smallest degree* of G : the second entry, with multiplicity, of the degree sequence of G sorted in nondecreasing order. Thus $\delta'(G)$ equals the minimum degree $\delta(G)$ whenever at least two vertices attain the minimum degree. A *leaf* is a vertex of degree one.

Conjecture 143 of *Written on the Wall II*, dated 2005, reads as follows [4]: *if G is connected and not a tree, then $t(G) \geq (g(G)+1)/\delta'(G)$* . At the time of writing, West’s current entry contains no solution note [4], and the conjecture is stated as a research-open item in the upstream Google DeepMind *Formal Conjectures* repository [9, 13]. We prove it in the equivalent, denominator-free form.

Theorem 4.1. *Let G be a finite simple connected graph that is not a tree. Then*

$$t(G)\delta'(G) \geq g(G) + 1.$$

The proof splits on $\delta'(G)$. When $\delta'(G) \geq 2$, deleting one vertex of a shortest (hence chordless) cycle leaves an induced path on $g(G) - 1$ vertices, and the inequality follows from $g(G) \geq 3$ by arithmetic. The substance of the theorem is the case $\delta'(G) = 1$, in which G has at least two leaves; here we prove the following lemma, which may be of independent interest.

Lemma 4.2 (Two-leaf lemma). *Let G be a finite simple connected graph that contains a cycle and has at least two vertices of degree one. Then G has an induced tree on at least $g(G) + 1$ vertices.*

The lemma is proved by a maximality argument: among the induced trees containing two prescribed leaves, a maximum-order one admits an outside vertex with two neighbours on the tree, which closes a cycle avoiding both leaves. Both the lemma and the theorem are sharp for every girth (Section 4.5).

Related work. The closest antecedent we are aware of is the induced-*forest* bound of DeLaViña and Waller [7] (see also the discussion in Hertz–Marcotte–Schindl [8]): if $f(G)$ denotes the largest order of an induced forest and $f_1(G)$ the number of leaves of a connected graph G with a cycle, then $f(G) \geq g(G) + f_1(G) - 1$. With two leaves this produces an induced forest on $g(G) + 1$ vertices, but the forest need not be connected, and the general forest-to-tree transfer of [8, Theorem 2.3] is too lossy to recover Lemma 4.2 from it. A targeted search of the foundational paper [6] found bounds for $t(G)$ in terms of order, size, radius, independence and clique numbers, but no bound involving girth or the second-smallest degree.

Concurrent formal resolution and priority. The current registers carry no solution note for Conjecture 143 [4, 13], and targeted formula and citation searches around [6, 7, 8] located no published proof of the theorem or of Lemma 4.2. These negative searches are not proof of absence. Moreover, on July 16, 2026, one day before the present proof was found, pull request

#4442 against the Formal Conjectures repository announced a machine-assisted resolution via an externally hosted Lean development [14]. The pull request remains unmerged at the time of writing, but we compiled the linked development against the repository’s pinned toolchain and confirmed that it proves the repository statement. Accordingly, we make no claim of priority for resolving the conjecture. The elementary proof and formalization presented here were obtained independently; because the argument is short and close in spirit to the 2004 forest bound, we also do not claim that it was never observed before.

Verification. Beyond the human-readable proof below, Theorem 4.1 has been formalized and machine-checked in Lean 4 [18] on top of Mathlib [19], against the pre-existing formal statement of Conjecture 143 in the Formal Conjectures repository [13]; see Section 4.6. As an independent falsification test, the theorem, the lemma, and the constrained form of the maximality argument were checked by two separately written exact checkers over the same atlas of all 1252 nonempty unlabeled graphs of order at most seven. This includes all 971 connected cyclic graphs in the atlas, and the constrained check covered all 199 unordered leaf pairs in the relevant graphs. No violation was found; the computation plays no role in the proof.

4.2 Notation

All graphs are finite and simple. For $S \subseteq V(G)$ we write $G[S]$ for the induced subgraph. A set S *induces a tree* if $G[S]$ is connected and acyclic, and

$$t(G) = \max\{|S| : G[S] \text{ is a tree}\}.$$

A path or cycle *in* G is always a subgraph; a path P is *induced* if $G[V(P)] = P$. The girth $g(G)$ of a graph containing a cycle is the minimum length of a cycle of G ; note $g(G) \geq 3$. The degree sequence of G is the multiset of vertex degrees sorted in nondecreasing order $d_1 \leq d_2 \leq \dots \leq d_n$, and $\delta'(G) = d_2$. This with-multiplicity reading of “second-smallest degree” is the one fixed by the Graffiti.pc definition list (Written on the Wall II, definition entry 65) [3], and it is the reading formalized in [13].

We use two elementary facts. First, in a connected graph on at least two vertices every degree is positive; hence if $\delta'(G) = d_2 = 1$ then $d_1 = 1$ as well, so G has at least two leaves. Second, a shortest cycle C of G is chordless, i.e. $G[V(C)] = C$: a chord splits C into two cycles, each shorter than C .

4.3 The two-leaf lemma

Lemma 4.3 (Lemma 4.2, strengthened). *Let G be a finite simple connected graph that contains a cycle and has two distinct vertices x, y of degree one. Then G has an induced tree on at least $g(G) + 1$ vertices; in fact some induced tree of order at least $g(G) + 1$ contains both x and y .*

Proof. A shortest x – y path in G is induced (a chord would shorten it), and it is a tree containing x and y . Hence the family of vertex sets

$$\mathcal{T} = \{S \subseteq V(G) : x, y \in S, G[S] \text{ is a tree}\}$$

is nonempty, and since G is finite we may choose $S \in \mathcal{T}$ of maximum cardinality. Write $T = G[S]$.

The set S is a proper subset of $V(G)$: if $S = V(G)$, then $G = T$ would be a tree, contrary to the assumption that G contains a cycle. Since G is connected and S is nonempty and proper, some edge of G joins S to its complement; let $z \notin S$ be a vertex with a neighbour in S .

We claim z has at least two neighbours in S . Otherwise it has exactly one, say a , and then $G[S \cup \{z\}]$ is the tree T with the single pendant vertex z attached at a : it is connected, and acyclic because every cycle of $G[S \cup \{z\}]$ through z would need two distinct neighbours of z in S , while a cycle avoiding z would lie in the tree T . Thus $S \cup \{z\} \in \mathcal{T}$ has larger cardinality than S , contradicting maximality.

Choose distinct neighbours $a, b \in S$ of z , and let P be the unique a - b path in the tree T . The edges of P together with za and zb form a (not necessarily induced) cycle of G of length $|V(P)| + 1$; possible further edges from z to $V(P)$ are irrelevant to its existence. Hence

$$g(G) \leq |V(P)| + 1. \quad (4.1)$$

Finally, neither x nor y lies on P . Indeed, every internal vertex of P has two distinct neighbours on P , and each endpoint (a or b) has one neighbour on P and the additional neighbour $z \notin S$; so every vertex of P has degree at least two in G , whereas $\deg_G(x) = \deg_G(y) = 1$. Since $V(P) \subseteq S$ and $x, y \in S$, the sets $V(P)$ and $\{x, y\}$ are disjoint subsets of S , so by (4.1)

$$t(G) \geq |S| \geq |V(P)| + 2 \geq g(G) + 1. \quad \square$$

4.4 Proof of Theorem 4.1

Proof of Theorem 4.1. Let G be connected and not a tree; then G contains a cycle, so $g(G) \geq 3$, and $|V(G)| \geq 3$, so all degrees are positive and $\delta'(G) \geq 1$.

Case $\delta'(G) \geq 2$. Let C be a shortest cycle of G ; as noted, C is chordless. Deleting one vertex of C leaves an induced path on $g(G) - 1$ vertices, which is an induced tree, so $t(G) \geq g(G) - 1$. Hence

$$t(G) \delta'(G) \geq 2(g(G) - 1) \geq g(G) + 1,$$

the last inequality being equivalent to $g(G) \geq 3$.

Case $\delta'(G) = 1$. Since all degrees are positive and the second entry of the sorted degree sequence equals one, the first entry equals one as well, so G has two distinct leaves. Lemma 4.3 gives $t(G) \geq g(G) + 1$, and multiplying by $\delta'(G) = 1$ finishes the proof. $\square$

Remark 4.4. The formal statement of Conjecture 143 in [13] quantifies over connected graphs on a finite vertex type with at least two vertices and positive δ' , without excluding trees, and uses the Mathlib convention that an acyclic graph has girth 0. That extension is immediate: for a connected tree G on $n \geq 2$ vertices, the whole vertex set induces a tree, so $t(G) \delta'(G) \geq n \geq 2 > 1 = g(G) + 1$. (Already $t(G) \geq 1$ suffices.)

4.5 Sharpness

Proposition 4.5. *For every $g \geq 3$ there is a connected non-tree graph G with $g(G) = g$, $\delta'(G) = 1$, and $t(G) = g + 1$. Moreover, for every $g \geq 3$ there is a connected non-tree graph H with $g(H) = g$, $\delta'(H) = 2$, and $t(H) = g - 1$, so the case split of the proof is tight as well.*

Proof. For G , take a cycle C_g and attach two pendant vertices (to arbitrary, not necessarily distinct, cycle vertices). Then $g(G) = g$ and $\delta'(G) = 1$. At most two cycle vertices support the pendants; deleting a cycle vertex supporting neither pendant leaves an induced subgraph on $g + 1$ vertices that is connected and acyclic, so $t(G) \geq g + 1$. No induced tree has $g + 2$ vertices, since the only induced subgraph on all $g + 2$ vertices is G itself, which contains a cycle. Hence $t(G) = g + 1$, attaining equality in Theorem 4.1 and Lemma 4.3.

For H , take the cycle C_g itself: $\delta'(H) = 2$, and the largest induced trees are the paths obtained by deleting one vertex, of order $g - 1$. At $g = 3$ this attains equality in Theorem 4.1, $t(H)\delta'(H) = 2 \cdot 2 = 4 = g(H) + 1$. $\square$

4.6 Formalization

The Google DeepMind *Formal Conjectures* project [9, 10] maintains Lean 4 formalizations of open conjectures; Conjecture 143 was added in June 2026 as the statement `conjecture143` in the file `FormalConjectures/WrittenOnTheWallIII/GraphConjecture143.lean`, marked `research open` [13]. The statement is the denominator-free real-valued inequality $g(G) + 1 \leq t(G) \cdot \delta'(G)$ for connected graphs on a finite vertex type with at least two vertices, under the hypothesis $\delta'(G) > 0$, with t and δ' given by the repository definitions `largestInducedTreeSize` and `secondSmallestDegree` and with Mathlib’s $\mathbb{N}$ -valued girth (which is 0 on acyclic graphs).

We have produced a complete, sorry-free Lean 4 proof of exactly this statement, machine-checked with Lean toolchain v4.27.0 against the Mathlib revision pinned by Formal Conjectures commit c252a41 [19]. The development follows the paper proof, with the supporting results proved as reusable graph-theoretic API (`SimpleGraph` namespace): the one-vertex extension of an induced tree along a unique neighbour (`IsTree.induce_insert_of_unique_adj`); induced trees from geodesics (`Walk.induce_support_isTree_of_length_eq_dist`) and the induced path obtained from a shortest cycle (`girth_sub_one_le_largestInducedTreeSize`); the existence of a maximum-order induced tree containing two prescribed vertices (`exists_maximum_induced_tree_containing`); the boundary-vertex and two-neighbour maximality arguments (`Connected.exists_adj_finset_compl`, `exists_two_adj_of_maximum_induced_tree_containing`); the cycle-closing certificate with the counting steps of Lemma 4.3 (`IsTree.girth_add_one_le_card_of_two_leaves_of_two_adj`, `girth_add_one_le_largestInducedTreeSize_of_two_leaves`); and the extraction of two leaves from $\delta'(G) = 1$ (`exists_distinct_degree_one_of_secondSmallestDegree_eq_one`). The main theorem is then assembled by the case split of Theorem 4.1 in under thirty lines. The proof uses no `native_decide` and no project-specific axioms. Auditing `#print axioms conjecture143` reports exactly `propext`, `Classical.choice`, and `Quot.sound`. The proof source is publicly archived in the author’s fork [17]. The ancillary files reproduce the branch after the metadata-only follow-up commit eb6445545bf839436642b895ad178cb397164dea.

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